

Chapter 22

Jabodetabek Megacity: From City Development Toward Urban Complex Management System

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Abstract Jabodetabek has developed from the small and separated city regions into a larger and unified megacity. This area consists of Jakarta as the core city and its surrounding areas namely Bogor, Depok, Tangerang and Bekasi. Recently Jabodetabek has become the largest megacity in Indonesia and plays the most important role within social, economic and political aspects. However, lack of planning capacity to deal with growing complexity in managing this area should be seriously remarked. The objective of this paper is to describe the dynamics of Jabodetabek starting from the early city development toward the future megacity development. It is divided into three sections comprising planning and development history; recent status of social-economic and physical-environmental situation; and future challenges that should be anticipated to achieve sustainable development. Obviously, planning and development strategy need to be transformed into adaptive, inclusive and integrated approaches within a continues incremental process to reach the development vision.

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22.1 Introduction

According to 2010s Population Census, Indonesia's total urban population is estimated to have reached more than 118 million people or about 50 % of the national total population. As in most developing countries, urbanization in Indonesia is characterized by heavy concentrations of urban population only in a few large cities. This tendency has been growing at an increasingly faster rate in the globalization era. The urban areas, especially large cities and megacities offer infrastructures and facilities as well as access to capital, labor and market for secondary and tertiary economic activities. The megacities have developed as concentrated economic activities, capital, and people facilitated by relatively better transportation and communication technologies.

Jabodetabek region is the largest urban concentration in Indonesia involving several administrative regions, it consists of three major provinces: Capital City of Jakarta, Banten (Tangerang Regency, Tangerang Municipality and South Tangerang Municipality) and West Java Province (Regency and Municipality of Bogor, Regency and Municipality of Bekasi, and Depok Municipality) with Jakarta as its center and Bodetabek (Bogor, Depok, Tangerang and Bekasi) as its suburban areas (*hinterland*) (see Fig. 22.1). Jabodetabek has a strategic role in the national development, particularly the economic, political, and socio-cultural structures. The growing socioeconomic concentration and activities in Jakarta and its surrounding areas have attracted migrants, particularly those from rural areas.

In spite of the conceptual unity, Jakarta and the surrounding regencies and municipalities governments have found it extremely hard to carry out well-coordinated policies as urbanized areas have expanded farther beyond the environs

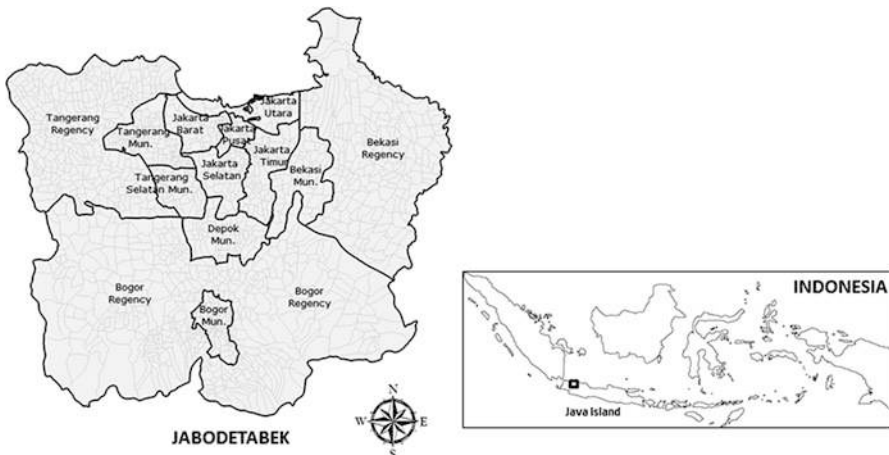


Fig. 22.1 Administrative Map of Jabodetabek

of Jabotabek following the expansion of the highway network. Due to suburbanization, Jabodetabek region is experiencing the growth of fringe areas which is more rapid than the core of the region and as a lifestyle involving a daily commuting to job places in Jakarta City.

Being significantly different from the urbanization and suburbanization process in USA and Europe, studies on Asian countries have stressed that the continuing outward expansion of the biggest metropolitan region has eroded the longstanding distinction between rural and urban (McGee 1991). The urbanization in east and south of Asian countries is characterized by the blurred distinction between 'rural' and 'urban' (McGee 1991, 1994, 1995; McGee and Robinson 1995). Some urban characteristics appear in rural areas, both physically and socioeconomically (Firman and Dharmapatni 1994; Firman 1997, 2003). Both agricultural and non-agricultural activities take place side by side in the adjacent areas of the urban centers, while the urban physical development extends beyond city administrative boundaries.

The trend of urban transformation in Jabodetabek has been driven by economic expansion such industrial complex and new satellite towns, and it has resulted in extended areas of mixed land use of city peripheries (Rustiadi and Kitamura 1998; Rustiadi et al. 1999; Rustiadi and Panuju 2000, 2002). By employing spatial statistics, Rustiadi et al. (1999) showed the land use mixture in suburban areas of Jabodetabek Region as well as in Tokyo Metropolitan suburbs (Rustiadi and Panuju 2000). McGee (1995) labels this phenomenon 'mega-urbanization' (Lin 1994; McGee 1995; Jones 2002), whereas in his earlier work he calls this phenomenon 'Kotadesasi' a phrase coined from the Indonesian language (Bahasa Indonesia) means the process of socioeconomic and physical integration between urban areas (Kota) and rural areas (Desa) (McGee 1991). Aguilar and Ward (2002) also identified similar phenomenon in Latin America. Despite McGee's "desakota" concept, many scholars have tried to develop various terms to describe the phenomena related to this region-based urbanization such as "extended metropolitan region" (McGee and Greenberg 1992), "mega urban regions" (McGee and Robinson 1995), "global city-regions" (Scott 2001), and "megacities".

Furthermore, Ginsburg et al. (1991) explained the region-based urbanization in East Asia and Southeast Asia, where large cities are often located in wet-rice production areas. Due to its labor-intensive nature of wet-rice agriculture, even before the period of industrialization, it had been common in these areas to have a population density almost equal to the cities in pre-industrialized Europe. The labor demand of wet-rice farming fluctuates from season to season, and this has prompted farmers to engage in various non-agricultural activities, such as petty trade and small-scale industry (McGee 1991). Various observers have pointed out that urbanization in Indonesia as well as other Southeast and East Asia countries are not centripetal, but rather proceeds as the formation of vast city-regions.

As a functional mega urban system, Jabodetabek region is still undergoing physical growth and the development of various activities. Various interests and rapid development in Jabodetabek have great impact on the spatial aspects and decreased carrying capacity of the environment. The results of the study by Rustiadi

et al. (2009) showed that the carrying capacity of the environment, especially land and water in Java Island, where Jabodetabek lies, is already overshot.

Spatial pattern in Jabodetabek region is characterized by spatial inconsistencies that occurred between existing land use and spatial plan. This condition generally occurs in urban areas and their surrounding areas. Inconsistencies also occurred between spatial planning (RTRW) and land capability (as a proxy of carrying capacity). These inconsistencies have caused some problems, such as land conflict as well as environmental degradation which may impact on the occurrence of anthropogenic disasters (floods, landslides, and so on). The previous research stated that there have been land use inconsistencies in Jabodetabek (2001) by 8.50 % of the total area (Nurhasanah 2004) and in 2010, there were 10.21 % inconsistent actual land use (Rustiadi et al. 2012). It requires a significant breakthrough to improve the controlling system of spatial planning effectively and gear it into a more efficient, productive and competitive Global Megacity.

22.2 Planning History and Related Development Issues

Jabodetabek development path could not be separated from the development of Jakarta. After the independence, Jakarta development has been growing rapidly. Moreover, this development ran progressively after the end of the Dutch aggression on 27 December 1949. During this post-independence aggression, Yogyakarta became an Indonesia capital city. In 1960, Jakarta position as the capital city was strengthened by the issuance of Government Regulation (2/1961) and in 1964 the Law 10/1964 was published. Laws and regulations related to Jakarta had been issued again after the 1997–1998 Indonesia economic crisis, namely Law 34/1999 and Law 29/2007.

22.2.1 Jakarta Planning After Indonesia Independence

Jakarta metropolitan concept actually has been known since 1950. In this year, an initial report was published to introduce the idea of metropolitan regions and metropolitan concept under the name “Jakarta Raya”. This idea was in accordance with the vision of Soekarno (President of Indonesia at that time) to build the national capital. This concept was introduced by a committee group between Indonesian and Dutch planners under the Ministry of Public Works and Engineering. This idea and concept became more concrete in 1952 with the term ‘Concept Plan’ for Greater Jakarta. This plan was based on the garden city scheme and its four development principles. Spatial structure developed in this plan was a concentric development model where Freedom (*Merdeka*) Square became the center, surrounded by several highways and greenbelt as an urban development boundary (Silver 2008).

In 1959, after 2 years of process chaired by Kenneth A. Watts, the Jakarta detail plan was finally approved by the Municipality of Greater Jakarta. This plan was known as the 'Outline Plan' which had a better detail rate than the previous plan. This plan also used better statistical data and information in the process. Based on the calculations made by the consultant, this plan required huge fund which cannot be fulfilled by the public development funds. As a consequence, most of the housing construction was suggested to be provided by private developers. In addition, as an in-between strategy, temporary housing was allowed to accommodate the rapidly growing population. Furthermore, Jakarta was inhabited by so many migrants from rural areas which still had strong roots and culture of rural lifestyle. This migration issue became a serious problem and attention for urban planners and academics. Migration had increased rapidly especially when the Japanese started to occupy Indonesia in 1942. Attempts to carry out the reduction of migration by creating more attractiveness in the origin area in the 1950s became one of the discourse raised by the University of Indonesia academic—H. J. Heeren. The combination of housing policy, the lack of infrastructure services, and the inability to resist the flow of migrants later became the cause of slums and squatter mushrooming in Jakarta. The satellite towns concept to accommodate residential areas in the future development was also discussed in this plan. Tangerang, Bekasi, and Bogor were identified as a satellite towns concept to avoid the overcrowding population in Jakarta (Silver 2008).

Although at that time the Outline Plan was made as good as possible, its implementation was constrained by the intervention of President Soekarno who did not like to be regulated by his servant and the plan made by it. One of the modern plan produced and had got rejection from President Soekarno, because of previous reason, was the development of electric tram which would connect Djatinegara to Matraman and Kramat to Senen (Silver 2008). In 1960, Kenneth A. Watts was also proposing to develop the existing towns surrounding Jakarta, in particular Bekasi to the east, Tangerang to the west, and Bogor to the south as an alternative to overcome overpopulation in Jakarta. To improve connectivity with these growth poles, the construction of the road network was needed (Silver 2008).

22.2.2 The Modern Metropolis and Master Plan 1965–1985

Recommendations contained in the Outline Plan and various consultations conducted by Kenneth A. Watts became the basis for making more detailed and practical development schemes, most notably the Master Plan 1965–1985 (Silver 2008). The Master Plan also marked a new Indonesian development era of Soeharto as Indonesia's second president who had replaced Soekarno. Furthermore, Jakarta had a new governor, Ali Sadikin, who became a part of history because of his strength of leadership. The Master Plan also confirmed Law 10/1964 regarding the position of Jakarta as the capital city of Indonesia. As the capital, this Master Plan was also made so that Jakarta, as the capital, has a high standard of international city.

Master Plan 1965–1985 also confirmed that all issues which hit Jakarta were mostly due to land use and control problem. Much of the land was controlled by government agencies acting on their own without the agreement of Jakarta leader. After the Master Plan was officially approved, further analysis was done based on the assumption that the construction of Jakarta should consider a regional perspective. The result was the need to create an integral plan between Jakarta and its surrounding cities forming Jabotabek (Jakarta, Bogor, Tangerang, Bekasi). However, compared with the concept of Jakarta and satellite cities planned in the Outline Plan, the Master Plan 1965–1985 had more emphasis on strategies that respond to local needs. If Ali Sadikin focused on local problems of Jakarta, Indonesia's United-Nations-financed consultants who helped prepare the Outline Plan and Master Plan 1965–1985 still held regional concept for future Jakarta development. The outer limits of the planning area were designated by Cikarang watershed to the east, Cisadane watershed to the west, and the mountain range to the south of Bogor (Silver 2008).

The planning concept of Jabotabek formally started in 1973 by the Ministry of Public Works, assisted by Dutch planners. One of the most important considerations in the preparation of an effective plan is the accurate data and the direction of *kampung* development in Jakarta. *Kampungs* in this perspective could not be treated as temporary slums, but they were permanent settlement depicting rural life to urban transition. One of the principles in this plan scheme was to use the railway lines to determine the development path of new settlement clusters. One such cluster was Depok which was projected to accommodate the overspill from the medium-low income residents of Jakarta. The Master Plan also planned protected areas because it has crucial protective function as the area between Bogor and Cibinong as the source of water, agricultural land in the north Bekasi, the east of Jakarta, the north and west of Tangerang which was also destined for the airport. The chairman of Dutch consultant team, L. Giebels, proposed to establish an organization to implement the Jabotabek Plan. The term of Jabotabek was also introduced by L. Giebels (Suselo 2003). In 1976 (Presidential Instruction 13/1976), Indonesia's government established Jabotabek Cooperation Body (*Badan Kerja Sama Pembangunan* or BKSP) of the joint secretariat of the Government of DKI Jakarta and West Java Province. In 1976, Jabotabek planning funding was taken over by the World Bank from the United Nations and Dutch consultants. In 1980, Jabotabek Metropolitan Development Plan (JMDP) was produced for the first time (Silver 2008).

22.2.3 *Urban Sprawl and the Rise of Botabek*

In 1983 when the Jabotabek team revised the 1980 plan, they kept their eyes open to the process of revised Jakarta 1985–2005 master plan which was also underway (Silver 2008). Development of public transport seemed to have no place in the 1980s until 1990s period. Dutch planners who had developed their vision in mass

transportation system connecting sub-urban development centers were considered too expensive. The Indonesian government and real estate developers chose to build housing in the suburbs by developing from highway network which had been built by President Sukarno in the 1960s. The completion of Jagorawi highway project in 1978 was the foundation stone of expansion of Jakarta toll road network system. In the following year, 1988, built-up area in Bogor, Tangerang, and Bekasi had reached 110,000 ha, or more than three times as much as built-up area in Jakarta. Large-scale residential projects—such as Bumi Serpong Damai—and those of medium-scale, combined with toll road network development and relatively cheaper land prices had made sub-urban areas surrounding Jakarta grow rapidly (Silver 2008).

In 1985 to 1999 period, the policy of the National Land Agency (BPN or *Badan Pertanahan Nasional*), which easily gave permission to build 80,000 ha of land in the outskirts of Jakarta provided incentives to cause urban sprawl. In Jakarta itself, land price was increasing rapidly and followed by the growing number of business complex developments. As a result of higher intensity of development in the elite area of Central Jakarta, many lower-middle income residents had been displaced. During the 1984–1994 period, the total population in Central Jakarta decreased by 15 % and became the only city in Jakarta which had experienced population decline. The growth of the property sector was also in correlation with Indonesia's economic growth in the early 1990s when Indonesia was classified as one of the Asian tigers (Winarso and Firman 2002; Silver 2008).

In addition to the property sector, manufacturing industry sector was also growing rapidly, especially in the outskirts of Jakarta. In 1985, Jabotabek produced 31 % of national industrial output, especially for textiles, electrical equipment, printing, publishing, and other chemical industry products (Firman and Dharmapatni 1994). The development of manufacturing industry in Jabotabek had gotten support from domestic and foreign investment, which the latest investment was largely dominated by Japan, South Korea, Hong Kong, and Singapore (Firman and Dharmapatni 1994).

As a consequence of uncontrolled development, environmental aspects had received negative impacts. Many farmers and their agricultural land were displaced due to construction of housing, industry, and other growing economic activities in Botabek. Physical development in this period often had not paid attention to environmental functions, for example development in Depok and Serpong which actually functioned as water recharge areas for Jakarta (Firman and Dharmapatni 1994).

22.2.4 Property Boom, Economic Crisis, and Its Recovery

The development of property sector which was increasingly out of control was suspected to be one of the main triggers of Indonesia's economic crisis in 1997 (Winarso and Firman 2002). The real demand of the property was actually not very

high; many investments were speculative and not based on actual need. This property development had surely sucked out a lot of loans from the banks and when the property sector collapsed, the financial sector was going to crash as its impact. Many infrastructures built in Jabotabek had high profit seeking motive, rather than delivering better services to the citizen. Economic crisis which hit in 1997 destroyed the booming property sector in Jabotabek.

The economic crisis had impacted on increasing abandon and idle land due to failed and uncompleted projects. Residential development in the periphery region has decreased significantly, followed by the industrial development in Bekasi and Tangerang. Many workers in Jakarta chose to go back to their hometowns due to being unemployed. Some migrants in Jakarta tried to take advantage of abandoned and idle land for urban agriculture activities (Indraprahasta 2013). The other phenomenon occurred was entering informal jobs such as street vendors and itinerant traders (Firman 1999). One year after the economic crisis, Indonesia experienced a change in the governmental and political structure, entering the regional autonomy era (Law 22/1999 and Law 25/1999). Economic development that was previously nuanced so national and centered around Jakarta had tried to expand by providing other municipalities (*kota*) and regencies (*kabupaten*) more authority to dig and utilize their own potentials. In the context of overall urban development in the 1998–2001 period, there was no significant progress of improvement (Firman 2002). This situation also applied to Jabotabek which run in accordance with its own recovery mechanism. Within this period, political aspect development had attracted more attention than economy. Directly and indirectly, changes in governmental structure had assisted Jabotabek in the recovery process.

22.2.5 *Jabodetabek in Democratic Era and Economy Acceleration*

Starting from 2004, the Indonesian governmental system has begun to find its ideal form, followed by the improving of national economic performance. This has a critical implication to the practice of spatial planning which had gained more attention and respect. Law 24/1992 which has been revised to Law 26/2007 was intended to strengthen spatial planning function in the development policy. As a result, the entire spatial planning in all regions of Indonesia should be revised in accordance with this new law, including Jakarta Spatial Plan. According to National Spatial Plan or RTRWN (Government Regulation 26/2008), Jabodetabek functions as the center of national activities in the national urban system. Depok municipality which has been continuously growing as an urban area had entered into the Jabodetabek acronym. In addition, the attention to the regional system of Jabodetabek as a whole has led Jabodetabek-Punjur (Puncak Cianjur) as a “national strategic area”. This area is planned to have two roles at once, namely as economic driver (Jabodetabek) and environmental protection (Punjur). To provide stronger direction of Jabodetabek-Punjur development, the Indonesian government had

made Jabodetabekpunjur Spatial Plan (Presidential Decree 54/2008). This period was also characterized by the demand to provide more participatory process as well as greater attention to environmental aspects. As an inherent complement of mainstreaming environment aspect in development policy, new environmental regulation requires all spatial planning documents to be examined through strategic environmental assessment or SEA (KLHS or *kajian lingkungan hidup strategis*) process (Law 32/2009).

The stronger and more robust decentralization and local autonomy practices as well as the improving performance of the national economy, development in Bodetabek find its enthusiasm again. There is also an indication of post-suburbia symptom experiencing in Jabodetabek (Hudalah and Firman 2012). In addition to continue-growing residential development, the manufacturing industry in the suburbs has also contributed increasingly to GDRP. The contribution of this sector in GDRP increased significantly from 24.6 % in 1985–1990 to 59.8 % in 2000–2005 and foreign direct investment (FDI) has played an important role in this increase trend (Hudalah and Firman 2012). Bekasi itself has emerged as a concentration of manufacturing industry development area in Jabodetabek. In the Asian level, Jababeka industrial zone (which is located in Bekasi) with an area of 5,600 ha become the largest industrial estate in Southeast Asia (Hudalah and Firman 2012).

The symptom of mega-urbanization also emerged between Jabodetabek and Bandung Metropolitan Area reflected from the formation of urban belt along 200 miles between Jakarta and Bandung (Firman 2009). This condition arose as a result of the opening of Cipularang (Cikampek, Purwakarta, Padalarang) toll road. This road has also encouraged the proliferation of industrial areas to the east of Jabodetabek, especially Karawang. The growing variety of industries in the region would threaten Karawang as a strategic region of food producers (paddy field) in Indonesia.

The current national policy is likely to facilitate economic capitalism in Jabodetabek, which can be reflected by the presence of the Master Plan of Acceleration and Expansion of Indonesia's Economic Development (*Master Plan Percepatan dan Perluasan Pembangunan Ekonomi Indonesia* or MP3EI) 2011–2015. MP3EI puts Jabodetabek as 1 of 22 major economic activities. Jabodetabek became 1 of 2 territorial-form activity of the 22 main economic activities. One other territory is a national strategic area (*kawasan strategis nasional*) of Sunda Strait; 20 other major economic activities are in the form of economic products.

22.3 Current Jabodetabek Megacity Development

22.3.1 Demography

Jabodetabek megacity covering an area of approximately 7,000 km², and had a total population of more than 26.7 million in 2010. The population of Jakarta city as the core of Jabodetabek was 8,502,818 and Bodetabek was 18,253,144 (Rustiadi et al. 2012). Jakarta reached its peak population growth in the periods of 1961–

1971 (55 %) and 1971–1981 (42 %). In those periods, the population increase in Jakarta was much greater than that of Bodetabek region. However, after 1981, the rate of population growth in Bodetabek has changed significantly and always been higher than that of Jakarta. In the period of 1981–1991, the population growth in Bodetabek and Jakarta were 68 % and 8 % respectively. This indicates that the population growth in Jabodetabek is no longer happening in Jakarta as the core, but it is distributed in Bodetabek region as its hinterland.

The highest population growth of Jakarta occurred in the period of 1961–1971. The highest population growth in the suburban areas of Jakarta (Bodetabek) happened in the period of 1981–1991. When the suburbanization was reaching its peak, the Jakarta population was already less than that of Bodetabek. Between 1991 and 2000 there was a decrease population growth in both Jakarta and Bodetabek. This was caused by the economic crisis in Asia, which also had an impact on the economy and population growth in Indonesia. However, in the period of 2000–2010 the number of population in Bodetabek increased again. It is necessary to examine this, which also becomes a major question: Is re-suburbanization happening again in Bodetabek in those periods.

Based on Rustiadi et al. (2012), the population share of Jabodetabek to the national population was increased from 6.10 % in 1961 to 11.26 % in 2010, but the proportion of the Jakarta city population to the total Jabodetabek population declined over time, from 49.1 % in 1961 to only 31.8 % in 2010. This reflects the suburbanization process in the peripheral areas.

22.3.2 Migration, Poverty and Informal Workers

Jabodetabek as the biggest economic center in Indonesia has powerful attractiveness that has pulled many people from other areas to come, live and work. Excessive in-migration has significantly contributed on rapid population growth and urbanization. Accelerating population growth has caused many problems related to providing sufficient infrastructure and maintaining the sustainability of the urban environment.

Based on the Central Bureau of Statistics (BPS), high in-migration not only happened in Jakarta but also in its surrounding area called Bodetabek. It could be seen in Table 22.1 that number of migrants who moved to Jabodetabek consist of

Table 22.1 Migration Characteristic in Jabodetabek

Area	Population number	Migrant from outside Jabodetabek	
		Number	%
Jakarta	9,556,049	1,427,933	14.94
Bodetabek	18,320,530	2,630,119	14.36
Jabodetabek	27,876,579	4,058,052	14.56

Source: Analysis on Population Census Data 2010 established by BPS

Table 22.2 Poverty Characteristic in Jabodetabek

Area	Population number	Poor population		Almost poor population		Vulnerable population	
		Number	%	Number	%	Number	%
Jakarta	9,557,798	475,008	4.97	728,604	7.62	1,203,612	12.59
Bodetabek	18,290,356	1,145,939	6.27	2,148,489	11.75	3,294,428	18.01
Jabodetabek	27,848,154	1,620,947	5.82	2,877,093	10.33	4,498,040	16.15

Source: Analysis on Integrated Basis Data for the Development of Social Security Program 2011 established by TPN2K

1,427,933 people who live in Jakarta and 2,630,119 people who live in Bodetabek. Although the number of migrants in Bodetabek is twice as Jakarta, the percentages are relatively similar. Clearly, the expansion of urban area beyond Jakarta has developed new pull factor that attracts more number of people.

In total, the number of migrants in Jabodetabek reached 4,058,052 people or 14.56 % of the total population. This number is relatively high and contribute significantly in the population growth. Instead, the number could be higher if the migrants that have lived less than 5 years, which are not defined as a migrant in BPS data, is also counted. Parallel with government plan to accelerate economic growth of Jabodetabek, the number of migrants will continue to increase in the future.

Numerous problems will increase if the pace of population growth persists. Various services including infrastructure, housing, education, health, job opportunities, water, energy, food, waste treatment, green open space, etc. will be difficult to provide. Therefore it should be anticipated by starting to develop the alternative urban development concept such as vertical growth, smart growth, compact city and so forth.

The other social problem in Jabodetabek is urban poverty. Along with the increasing share of urban inhabitants within the total population, the number of poor people in urban area is also growing. In addition, many poor people are interested to move to urban areas because high economic growth in the cities offers many alternative occupations within formal and informal sector.

In 2011, the National Task Force for Poverty Alleviation (*Tim Nasional Percepatan Penanggulangan Kemiskinan*—TNP2K) has established integrated basis data for the development of social security program. They collected data of 40 % population with the lowest prosperity. This population is divided into four categories including very poor, poor, almost poor and vulnerable poor based on a poverty line established by BPS in 2011 which is Rp 233,740/capita/month.

The first two categories include number of people who get income below the poverty line, while the rest include number of people who get income slightly above the poverty line. Within this paper, all of vulnerable people were only divided into two groups comprising poor population (below the poverty line) and almost poor population (slightly above the poverty line).

It could be seen in Table 22.2 that the number of poor people just 4.97 % of the total population in Jakarta and 6.27 % of the total population in Bodetabek. However, the number of almost poor people is higher which reached 7.62 % of the total population in Jakarta and 11.75 % of the total population in Bodetabek.

It showed that many people will be easily fall down to the below poverty line if there is a shock within the economic system.

Afterward, the number of vulnerable population in Bodetabek is higher than Jakarta. It showed that accumulation of vulnerable people mostly happened in the urban fringe of Jakarta. This area is characterized by mix urban-rural livelihoods, therefore seemingly the conflict between urban and rural has lessened prosperity of the people.

In total, the number of vulnerable population in Jabodetabek is relatively high which reached 4,498,040 people or 16.15 % of the total population. Thus if the economic system becomes unstable, the number of people who live below the poverty line will become threefold. It should be anticipated because there are a lot of global–local issues that will threaten Jabodetabek social-economic system in the future.

The next social problem that become important issue is the existence of the informal sector. The informal activities have become the easier entrance for migrants to get involved in urban economic activities and earn any income, and also important to support formal workers that have limited wage. The competitiveness of Indonesian economic that rely on low wage labor actually is supported by the existence of the informal sector that could provide various cheap products.

Actually the problem of the informal sector is not related to its existence but how to minimize its negative impact on Jabodetabek development. It is known that the informal sector in developing countries is disorganized, thus it produces numerous negative externalities such as congestion, waste, slum areas, crime, etc. It could be seen in Table 22.3 that the informal workers in Jabodetabek could be divided into 4 groups comprising self employed; self employed with temporary workers; free worker; and unpaid worker. Self employed is a person who has own business or offer a particular skill such as tailor, carpenter, builder, serviceman, etc. Self employed with temporary workers is a self employed who is helped by impermanent worker or sometimes unpaid worker. Free worker is a person who doesn't have permanent employer. Unpaid worker is a person who works without payment and usually this person work on his/her family.

Informal workers in Jabodetabek were dominated by self employed which reached 22.22 % of the total labor. It was followed by free workers 7.74 %, unpaid workers 2.01 %, and self employed with temporary workers 1.25 %. In total the number of informal workers in Jabodetabek reached 33.22 % of total labor. This number is quite high since one third of labor in Jabodetabek work in the informal sector.

In general, the number of informal workers in Bodetabek is higher than Jakarta. It might be caused by the high pace of urbanization in this area whereby the interaction between urban and rural has opened more opportunities for the informal sector development. Still, it needs to be managed and organized in order to avoid chaotic development as well as negative externalities that could harm the public life.

Table 22.3 Informal Workers Characteristic in Jabodetabek

Area	Labor number	Self employed		Self employed with temporary workers		Free workers		Unpaid Workers		Informal workers	
		Number	%	Number	%	Number	%	Number	%	Number	%
Jakarta	4,285,981	849,563	19.82	46,704	1.09	248,386	5.80	77,310	1.80	1,221,963	28.51
Bodetabek	7,175,529	1,697,212	23.65	97,126	1.35	638,766	8.90	153,004	2.13	2,586,108	36.04
Jabodetabek	11,461,510	2,546,775	22.22	143,830	1.25	887,152	7.74	230,314	2.01	3,808,071	33.22

Source: Analysis on Population Census Data 2010 established by BPS

22.3.3 *Economy*

Jabodetabek Metropolitan Area becomes a center of national economic activities that are characterized by high population density and intensive spatial interaction with other regions. In national economy, Jabodetabek contributed 25.52 % of total national GDP in 2010, 17.92 % contributed by Jakarta city and 7.59 % contributed by Bodetabek (Rustiadi et al. 2012).

Based on the contribution of each sector to the total GDP of Java Island in 2010, Jabodetabek contributed 76.8 % for finance, 60.7 % for construction and building, and 56.6 % for transportation (Rustiadi et al. 2012). Other sectors which also had significant contribution were industrial sector; government services; electricity, gas and water; as well as trade, hotel and restaurants with the percentages respectively 43.8, 41.6, 41.3 and 36.5 %. The other two sectors which are agriculture and mining had small contribution, respectively 3.7 and 5.8 %. Jabodetabek's GDP contributed around 42.8 % to the total GDP of Java Island.

In terms of number of employees which works in each sector, it indicates that the financial sector as the biggest contributor to GDP also contributed most to employment at around 47 %. On the other hand, other sectors with a quite high number of employment are electricity, gas and water (38 %); transportation (31.2 %); services (27.8 %); trade, hotel and restaurants (27.6 %); industries (27 %) and other sectors (39.5 %).

Related to the productivity defined as the comparison of GDP to the total employment, it shows that Jabodetabek had the productivity of Rp 48.55 million/worker per year, whereas non Jabodetabek region (*rest of Java*) had the productivity of Rp 15 million/worker per year. This shows that high productivity in terms of GDP in Jabodetabek area is greater than the rest of Java.

In the Jabodetabek itself, there were three dominant sectors which have high contribution to the total Jabodetabek's GDP comprising: industrial sector (28.36 %), financial sector (20.66 %) and also trade, hotel and restaurant sectors (20.24 %).

22.3.4 *Land Use/Cover Change (Built-Up Area Development)*

Studies on land use change including urbanization have regarded a region as either an aggregated (lumped) system or a disaggregated (distributed) system. The aggregated system is used to estimate only the amount of the land use change considering the region as a mass (Hansen 1959; Mfungahema and Kitamura 1998; Rustiadi and Kitamura 1998; Wang and Zhou 1999), while the disaggregated one aims to simulate the land use change amount as well as the spatial distribution of land use types (Wu 1998; Yeh and Li 1998).

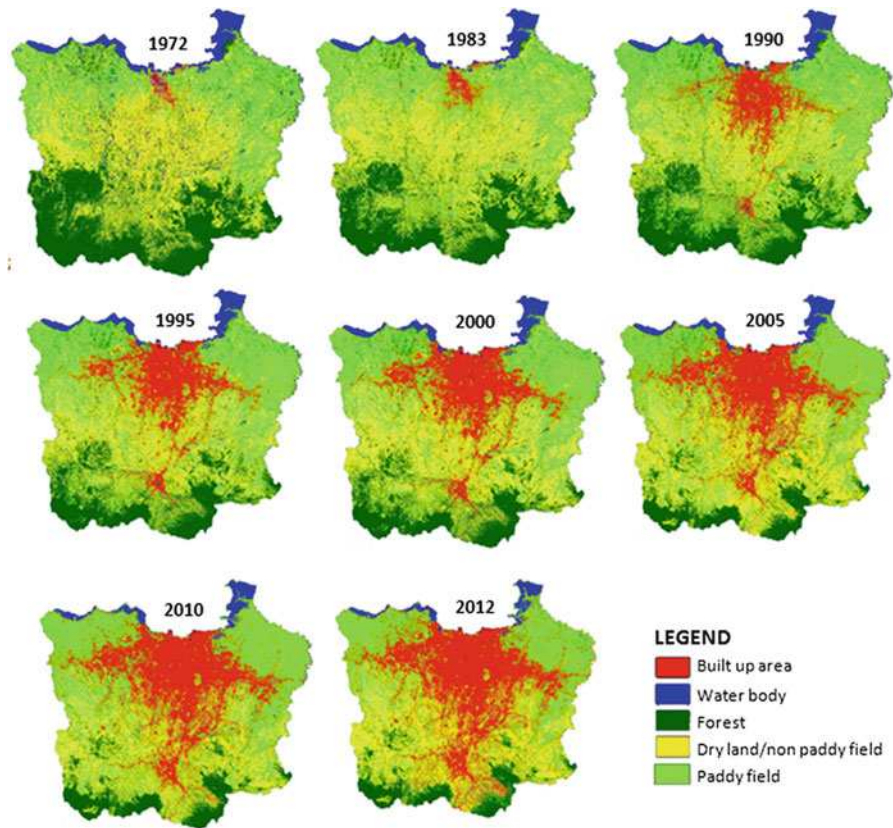


Fig. 22.2 Map of land use/cover in Jabodetabek in 1972–2012

Land use change is generally affected by both natural factors, such as geographical and soil characteristics, and socioeconomic factors, such as population growth, development plans, land use control, zoning and other related laws. Land use changes in Jabodetabek were dominantly caused by land conversion into the concentrated urban activities. Land conversion in this region tends to increase every year. In Jabodetabek, many fertile farmlands and other greenery areas converted to built up areas (settlement, industries, commercial services and others). The resulting images of land use/cover changes (LUCC) from 1972 to 2012 are presented in Fig. 22.2.

In 1972, the built-up area in Jabodetabek was still very limited (around 6,500 ha) and most of them were concentrated in Jakarta. On the other hand, we found some forest area distributed in Bogor and Depok. In the same year, land dominantly used for dry land (non paddy field) approximately 230,000 ha. In 1983, built up area was increased from 6,500 ha (1972) to 12,000 ha. We found that some part of the forest areas converted into paddy field and dry land (non paddy field), so the paddy field

and dry land were increased respectively from 173,000 and 230,000 ha to 215,000 and 235,000 ha. Based on land use/cover map of Jabodetabek in 1990, it can be seen that built-up area in Jakarta was getting wider. This phenomenon was also happening in Bogor, Tangerang, and Bekasi. Most of urban expansion in those areas during that period come from the conversion of dry lands and forest areas. Then, for more than 20 years, the built-up area in Jabodetabek was increasing as well as its distribution. The urban expansion spread out especially along the major highways.

Rapid land conversion in Jabodetabek occurred due to several factors such as the increasing number of population, which requires more space for housing and facilities development. From 1995 until 2012, built up area in Jabodetabek was growing rapidly. During those periods, urban area in Bogor, Depok, Tangerang and Bekasi were also expanding. During that time, almost the whole area of Jakarta was urbanized. Many hectares of forest area, especially in Bogor were converted and tends to decrease over the time.

For more than two decades, the utilization of built-up area was no longer controllable. The development of housing and industrial areas was growing more rapidly. Most of the dry land and paddy field in Bekasi and Tangerang were also changed into built up areas. In Depok were previously dominated by dry land had turned into settlement and industrial areas by 2010.

The land conversion from paddy field and dry land to built-up areas has had a significant impact on the condition of the surrounding areas. Since the expansion of urban area, the land conversion for housing and built-up area continues to increase. The land use of the forest area was falling significantly. Approximately 71 % or 152,000 ha of forest have disappeared from Jabodetabek through land conversion during 1972–2012. Dry land has also reduced by 31 % or 71,000 ha, whereas the built up area has expanded 31-folds from that of 1972 (increased around 195,000 ha during the same period). This reflects a rapid conversion rate over time.

22.3.5 *Environmental Problems*

The suburbanization has led to the conversion of primary agricultural land and green open space (also including forest area) and created environmental problems as a result of the land conversion without considering the function, capacity and suitability of the land itself. Environmental damage was getting worse in Jabodetabek over the time as shown in some studies which analyzed the environmental carrying capacity of Jabodetabek and Java island (KLH 2007; Rustiadi et al. 2009, 2010).

The real impact of the overshoot carrying capacity is the increasing anthropogenic disasters (such as flood and landslides) in the Jabodetabek region in terms of quantity, intensity and distribution. Figures 22.3 and 22.4 showed respectively the distribution maps of flooding and landslides occurred in each village of Jabodetabek in 2000, 2003, 2006, 2008, and 2011. The data were obtained from village potential statistic data (PODES) established by the Central Statistic Agency (BPS).

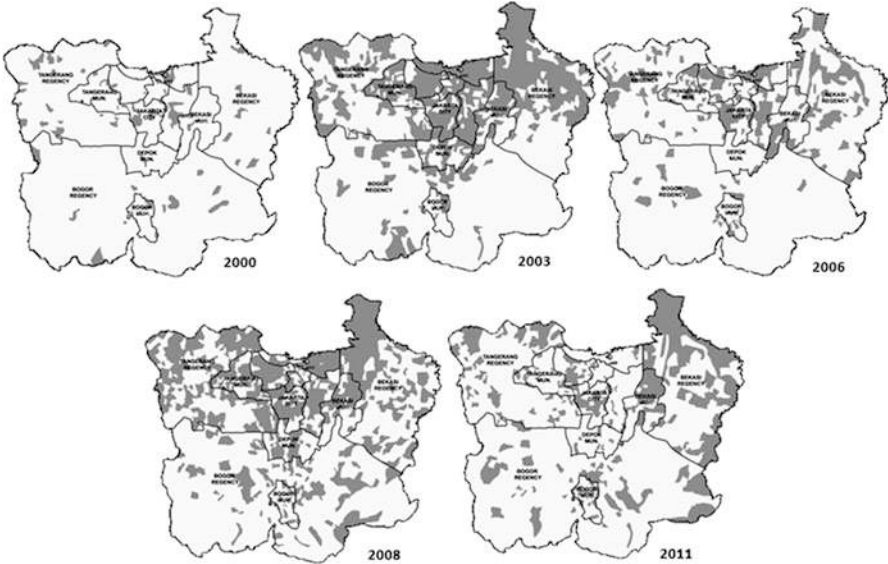


Fig. 22.3 Map of Flood Distribution in Jabodetabek. Based on PODES Data in 2000–2011

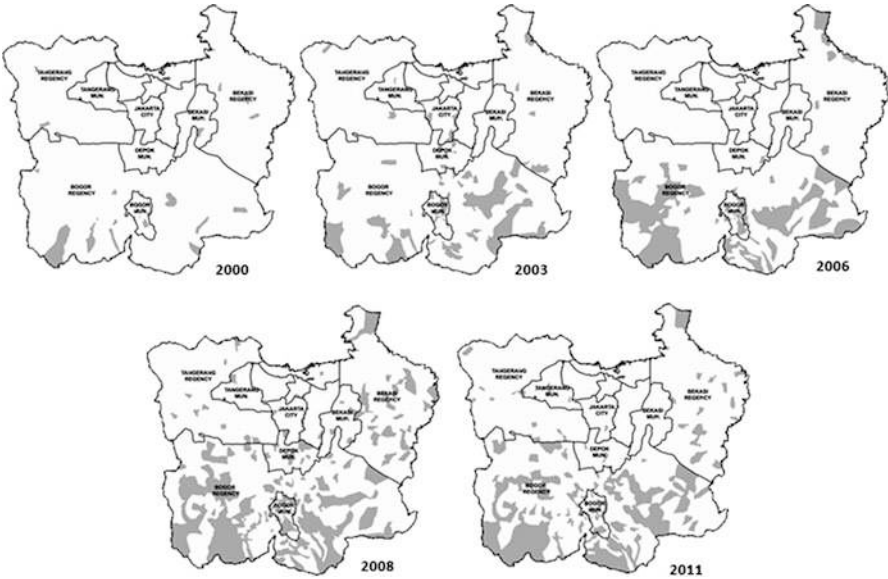


Fig. 22.4 Map of Landslide Distribution in Jabodetabek. Based on PODES Data in 2000–2011

In 2000, the number of villages experiencing floods was 102 or around 6 % of the total number of villages. In 2003, the number goes up, involving 609 villages (around 35.9 %). Then, in 2006 and 2008, the percentage of villages struck by floods respectively were around 19.5 and 37.9 %. Afterward, in 2011, the number of villages experiencing floods was only 338 (22.6 %). This pattern indicated the 5 years cycle of floods whereby in 2003 and 2008 the number of villages hit by flood increased.

Meanwhile, the number and percentage of villages experiencing landslides in Jabodetabek from 2000 to 2008 were also increasing over the time. It can be seen that the number of villages experiencing landslides in 2000 was only 26 (or around 1.53 %). But during 11 years (from 2000 to 2011), the number of villages and the percentage tends to increase, i.e. reaching 95 villages (5.6 %) in 2003, 134 villages (7.9 %) in 2006, 305 villages (17.94 %) in 2008, and 170 villages (11.37 %) in 2011.

The environmental issues and anthropogenic disasters are not merely caused by technical aspects but also non-technical such as institutional issues in which land control issues are included. The capacity of a planning system is determined by the institutional capacity to control the natural resource exploitation. The government role is much needed in this case. Unfortunately, a discrepancy between regulation, policies and planning concerning the current exploitation or utilization oftenly occurs.

Essentially, the frequent incidences of anthropogenic disasters are caused by the inconsistency in spatial planning as a result of some factors for instance: (1) the existing land use is not complying with the existing spatial planning; and (2) spatial planning is not in line with the regulations/principles. In addition, land use change provides many economic and social benefits, but comes at a substantial economic cost to society. Land conservation is a critical element in achieving long-term economic growth and sustainable development. Land use policy, however, must strike a balance between private property rights and the public interest.

22.4 Future Development of Jabodetabek Megacity as a Complex System

Jabodetabek has developed from the city region into megacity that exhibit a very complex system. Base on explanation in the previous section, the pace of urbanization in this area is very high and the urban area has expanded beyond the administrative boundaries. The connectivity among regency/municipality has become more intense and covers multidimensional facets comprising demographic, economic, social, cultural and environmental.

The regional complex system within Jabodetabek megacity is characterized by mutual interaction among subsystems which be present on a different scale, function, activities, and sub-region. Thus interaction among subsystems and its implication on managing Jabodetabek would be further explained in the following section.

22.4.1 *Global–Local Interaction*

Jabodetabek Metropolitan Area is initially developed in order to accelerate economic growth whereby the connectivity with the global market should be strengthened. It was expected to attract global investment that could stimulate the leap of economic growth. As occurred in other Asian cities, it has created numerous manufacturing industries based on the integration of global capital and region's cheap labor, then it develops a regional value chain of supply, production and sales (Dahiya 2012). Accordingly, by mid-1990s the development of this area appeared to be heading toward global city status (Firman 1999).

Dazzled by tremendous economic growth, raising of various issues within the local scope seemingly is neglected by the government. Excessive in-migration, raising informal activity, increasing urban poverty have impacted on congestion, emerging slum areas, widening income disparity, etc. While rapid urbanization as a result of economic growth has caused natural and agricultural land conversion, food insecurity, environmental degradation, etc.

Finally the contact between global and local is inevitable. Because they are often conflicting each other, then it has caused many troubles. Base on a global perspective, worsen local situation have made a lot of difficulties to increase their productivity and efficiency. Whereas, most of the local people against global economic activities that do not give significant impact on increasing local prosperity.

Surprisingly, nowadays local issues have grown up toward a global scale, whereas global capital started to expand their business to the local scope. Many city governments, NGOs and world aid organizations have started to devote much attention to the improvement of local urban life. Therefore some local issues such as congestion, urban poverty, slum area, environmental risk etc. have become common problems that should be resolved together. On the other hand the global economic network has developed their influence to the local level, although their role seems to give good and bad effects simultaneously. The initiative of global company to develop CSR (Corporate Social Responsibility) program has given a lot of benefit to the local people. However, the expansion of their business comprising the development of cheap and mass products, supermarket and mini-market network, and easier financial credit have threatened the existence of local economic activities.

Obviously, global–local interaction has become interrelated and more complicated than before. Recently, many global issues such as climate change, food insecurity and increasing price of energy are closely related to people daily life in Jabodetabek. On the other hand local issues such as increasing environmental risk, congestion, urban poverty, and informal worker have directly impacted on decreasing productivity and efficiency of global economic activities.

Unfortunately, planning capacity to deal with complicated global–local interaction is still limited. An integrated development policy that could put all of the good initiatives into planning is really required. Therefore all people or institutions that will participate in resolving local problems could fully understand what has been done and should be done. This kind of planning called as collaborative planning,

and it is more than just participatory planning that has been simplified into a public meeting and consultation.

Furthermore, the capability to recognize local economic activities should be increased, thus the objective of Jabodetabek to reach global economic competitiveness doesn't threat local economic development. Recent studies reveal that local and informal economic activity has a significant contribution in the urban economics in developing countries (Goethert 2013). Since many urban dweller also involved within this informal economic activity, then it could strengthen people prosperity as well as resilience of the city.

22.4.2 Economy-Ecological Interaction

Nowadays the role of Jabodetabek in the national economy is still dominant although the decentralization policy has been implemented since year 2000. Instead of distributing economic value, the economic polarization in Jabodetabek after decentralization era is even stronger than before. As described by Dahiya (2012) megacities in Asia act as dominant forces in both regional and global economies due to their important role in political power, financial capacity, and knowledge development.

The position of Jabodetabek has become more important since the last global economic crisis in 2008. Indonesia, India, and China are the three Asian countries which still had positive economic growth. In order to sustain the growth, the government intends to push the economic performance by focusing economic investment in the existing urban area especially Jabodetabek.

Along with the extending urbanization and growing population within Jabodetabek, many scholars have warned about the raising of environmental risk. The concern about environmental degradation has been acknowledged in a report by the United Nations Environment Program (UNEP) in 1983 (Douglass 1989). However, urbanization has continued to sprawl, thus the environmental degradation keeps accelerating and has turned into an environmental hazard. The most important hazard in Jabodetabek is a flood that has caused many losses including life and material (Douglass 2013).

Obviously, environmental aspect should be seriously considered in the planning. It should be connected and embedded into the economic development scenario. Unfortunately it is not easy to be done because the relationship between economic and ecological system within megacities has been developed into the more complicated interactions.

Regarding to this complexity, preventing environmental hazards through infrastructure engineering, spatial zoning or environmental valuation will be insufficient. Focus of planning instrument need to be directed to connectivity, circularity, and dynamics. Connectivity means developing connections between economic and ecological system. Circularity means how to manage flows between both systems

that has been connected. Dynamics means how to lead this connectivity and circularity into continual improvement toward a better system.

Related to the connectivity, some initiatives that bring nature into the urban life is really important. In the past paradigm, urban and nature has become unconnected because urban is regarded as a place for the development of artificial systems. The development of green open space which is provided not only for supporting human amenities but also conserving the life of local species is really interesting idea. Related to the circularity, urban waste management that promotes activities to reduce, reused, recycling and recovery could be seen as a case to develop circularity. It could be expanded into managing the cycle of energy, food, and water. Related to dynamics, the planning paradigm that view megacities as an organic system that has capabilities to learn, adapt, change and develop is really important. This will lead the connectivity and circularity toward achieving development visions. It is important for the megacities development which have to deal with various economic-ecological issues within global and local scale that always change time to time.

22.4.3 *Urban–Rural Interaction*

Urban–rural interaction in Jabodetabek has persisted for more than three decades since the emerging of *desakota* concept by McGee (1991). Actually, it had happened a long time before McGee recognized this area as a *desakota* region. Many scholars regarded this area as a specific feature within Asian megacities. Extending urbanization has penetrated this dense agricultural area and caused chaotic urban–rural land use as well as mixed urban–rural livelihoods.

Currently there is a strong debate concerning the future of *desakota* region. McGee (2010) argued that the existence of this area has become more important since the volatility of fuel and food price in the global market has threatened food security within megacity. It is related to the capability of this area to produce food as well as provide various ecosystem services. On the other hand, the development of industrial estates within peri-urban Jabodetabek has converted the chaotic land use into the planned and new cities, and this is believed as a sign of the end *desakota* region (Hudalah and Firman 2012).

Obviously, the development of industrial estates that neglect the important role of *desakota* region showed the absence of urban–rural linkage conception in a planning. Douglass (1998) has proposed the idea of developing urban–rural linkage policy in order to manage *desakota* region. It is focused on managing the flow of people, production, commodities, capital/income and information between urban and rural. In terms of spatial, it is implemented in the form of a hierarchical urban system comprising the mutual connectivity among big and middle cities within urban, and small cities within rural. Unfortunately, it failed to be implemented because a hierarchical urban system is hardly to define within mixed urban–rural livelihoods in *desakota* region.

Therefore up till now the urban policy still neglect the importance of rural area, whereas rural development only focus on developing on-farm activities. Within this situation, linking urban–rural system in Asian megacities still becomes a challenge that should be resolved. Instead of relying on the urban hierarchical system, a new approach is needed to manage the mixed urban–rural livelihood.

One of the solutions that could be offered is developing hybrid urban–rural development system within desakota region. Urban activities in desakota region will differ from urban activities in the urban area. It could be similar to the eco-village conception whereby modern life in this area is developed within an eco-friendly infrastructure. Transportation, energy, water, housing, and industrial development need to be designed properly in order to reduce resources used and produce a clean and healthy environment. It will become important to support the quality of natural and agricultural land adjacent to this urban area.

On the other hand, agricultural activities in desakota are different from rural agriculture because they should adapt to the urban system and urban environment. Therefore, intensive agriculture that usually uses chemical substance should be changed into organic farming that produce fresh and healthy products. Besides, the farming technique should be developed toward eco-friendly ways in order to increase the role of agriculture land in providing environmental services. Malaque and Yokohari (2007) argued that segmented farmland patches that remain in urban fringe areas could provide various services including the capacities for water retention capability, micro climate control, conservation of visual quality, and the supply of safe, fresh food. Surely, the farmer should get additional payment for the ecosystem services provided by their agricultural land.

22.4.4 Multiple Local Government Interaction

Lack of coordination among local government within Jabodetabek has become a classical issue that unresolved yet until now. Instead of developing integrated action, the policy has been fragmented in more numbers of provinces, municipality and regencies. At province level, the establishment of Banten Province in the year 2000, has separated Tangerang municipal and Tangerang district from West Java Province. Therefore Jabodetabek that previously comprising Jakarta and West Java Provinces has changed to an area that covers Jakarta, West Java and Banten Provinces. At municipality/regency level, Depok that previously was part of Bogor district has become a new municipal in the year 1999, while South Tangerang that previously was part of Tangerang Regency has become a new municipal in 2008.

The emerging of new province and municipals in Jabodetabek is related to the urbanization process. When certain area is urbanized and have sufficient income and population, then it will develop into a new municipal. Following this trend, government policy within Jabodetabek will become more fragmented in the future. It will exacerbate the problem because spatial discontinues of development policy will increase, even each municipal/regency will compete each other for getting higher local income.

Apparently, developing integrated governance within Jabodetabek is extremely difficult although most of stakeholders know that it is required to reach sustainable development. Many initiatives such as the formation of coordinating government institution and establishing particular President Regulation to manage this area have not given any significant effect. The idea to form a particular government institution for managing Jabodetabek region also has been proposed, but it is difficult to be realized because all local governments tend to against this idea.

Seemingly, relying on government initiatives in order to build multiple local government interaction will be ineffective. As suggested by Laquian (2008) a good policy should be pushed by the involvement of civil society groups, supported by social and mass media that enable citizen to get information and participate actively in public decision making. It means inclusive development needs to be developed to face the problem of fragmented governance in Jabodetabek.

In order to stimulate the involvement of the citizen, people concern about development policies should be strengthened by raising their capacity through the learning process and information dissemination. Jabodetabek as the center of knowledge with the advanced information technology has a capability to increase the capacity of people to participate in the public policy. As Laquian (2008) stated policies and strategies that achieve the welfare of the whole society will overcome the particular interest of specific groups including separated local government.

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